

Unequal Vulnerability: The Differential Impact of Label Flipping Attacks Across Classes

Pinlong Zhao
School of Cyberspace, Hangzhou
Dianzi University
Hangzhou, China
pinlongzhao@hdu.edu.cn

Mengyang Li
Tianjin Normal University
Tianjin, China
limengyang@tjnu.edu.cn

Pengfei Jiao
School of Cyberspace, Hangzhou
Dianzi University
Hangzhou, China
pjiao@hdu.edu.cn

Huijun Tang
School of Cyberspace, Hangzhou
Dianzi University
Hangzhou, China
tanghuijun@hdu.edu.cn

Ou Wu*
HIAS, University of Chinese Academy
of Sciences
Hangzhou, China
wuou@ucas.ac.cn

Abstract

Label flipping attacks stand as a potent and practical threat to the integrity of machine learning models. While extensive research has focused on designing sophisticated attack and defense mechanisms, the underlying factors that govern a model's susceptibility remain underexplored. This paper reveals a critical phenomenon: the impact of label flipping attacks is highly differential across classes, strongly correlated with the intrinsic confusability between the source and target classes. We provide a rigorous theoretical analysis, demonstrating that a lower standardized separation between classes fundamentally leads to greater vulnerability. Grounded in this insight, we propose Confusability-Aware Contrastive Learning (CACL), a targeted defense that maximizes the feature-space separation for the most vulnerable class pairs. Extensive experiments validate the strong link between class separability and vulnerability, and show that CACL significantly mitigates the attack's impact while providing superior protection for the most susceptible classes. Our code is at <https://github.com/Pinlong-Zhao/Unequal-Vulnerability>.

CCS Concepts

• Security and privacy; • Computing methodologies → Machine learning;

Keywords

Machine learning, Data poisoning, Label flipping attack, Unequal vulnerability

ACM Reference Format:

Pinlong Zhao, Mengyang Li, Pengfei Jiao, Huijun Tang, and Ou Wu. 2026. Unequal Vulnerability: The Differential Impact of Label Flipping Attacks Across Classes. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792713>

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License. WWW '26, Dubai, United Arab Emirates
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792713>

1 Introduction

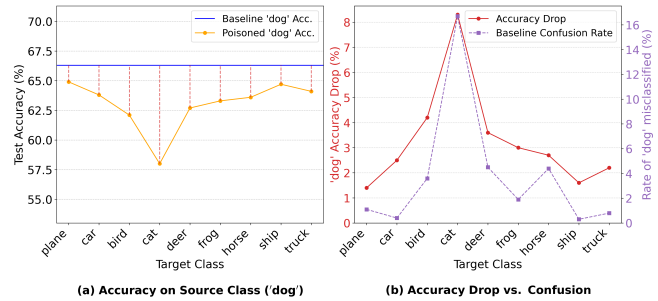


Figure 1: The differential impact of label flipping attacks. We poison the 'dog' class by flipping its labels to each of the other 9 classes in CIFAR-10. (a) The accuracy of the source class 'dog' before the attack (blue) and the accuracy after being poisoned towards each target class (orange). (b) The accuracy drop of the source class 'dog' (red) for each attack target, showing a large variance. This drop strongly correlates with the baseline model's confusion rate (purple).

The unprecedented success of deep learning has revolutionized numerous fields, from computer vision to natural language processing [32]. This success, however, is critically dependent on the availability of large-scale, high-quality training data. In many real-world scenarios, such as crowdsourcing [15, 38] or federated learning [20, 36], training data is aggregated from untrusted sources, making it a prime target for adversarial manipulation. Among various data poisoning strategies, label flipping attacks are particularly concerning due to their simplicity and effectiveness [5, 35, 39]. By maliciously altering the labels of a small fraction of the training data, an adversary can significantly degrade a model's performance or even implant backdoors, posing a serious threat to the deployment of reliable machine learning systems [6, 21].

Current research on label flipping attacks predominantly focuses on developing more potent attack strategies [8, 16, 34] or more robust general-purpose defenses [2, 3, 17, 26]. A common thread in these studies is the evaluation of a model's vulnerability through its average performance degradation across all classes. This global

perspective, however, obscures a crucial and more nuanced reality. In this work, we reveal that the vulnerability to label flipping attacks is not monolithic; instead, it exhibits a starkly unequal pattern across different classes. As illustrated in Figure 1, the damage inflicted by flipping labels from a single source class to various target classes varies dramatically. For instance, when poisoning the ‘dog’ class by flipping its labels to ‘cat’, the model’s accuracy on the ‘dog’ class plummets. In contrast, flipping ‘dog’ labels to ‘plane’ results in a far less severe impact. This differential impact suggests that a model’s susceptibility is not random but is systematically linked to the specific class pair involved in the attack. The clean model’s own confusion patterns hint at an underlying principle: attacks are most effective when they exploit the model’s pre-existing difficulties in distinguishing between certain classes.

This striking observation motivates the central question of our research: What fundamental properties govern this unequal vulnerability to label flipping attacks? We hypothesize that the efficacy of such an attack is deeply rooted in the intrinsic separability of the source and target classes. Classes that are already difficult to distinguish in the feature space provide natural fault lines for adversaries to exploit, amplifying the poison’s effect. An attack that pushes two already-close class clusters together is far more potent than one trying to merge two well-separated clusters. To formalize and address this, our work makes the following key contributions:

- We are the first to systematically identify and quantify the differential impact of label flipping attacks across classes, revealing that attack vulnerability is a structured, predictable phenomenon rather than a random one.
- We provide a rigorous theoretical framework that explains this unequal vulnerability, proving that it is fundamentally governed by the Standardized Separation between classes.
- Grounded in our theoretical insights, we propose a novel defense mechanism, Confusability-Aware Contrastive Learning (CACL), which hardens the model by selectively maximizing the feature-space separation of the most vulnerable class pairs.
- Through extensive experiments on both simulated and real-world datasets, we validate our theory and demonstrate that CACL not only enhances overall model robustness but also provides superior protection for the most susceptible classes, leading to a more equitable defense.

2 Related Work

Our research is positioned at the intersection of data poisoning and model vulnerability, focusing on a critical, overlooked aspect: why attacks have a differential impact across classes. We connect this unequal vulnerability to the intrinsic separability between class representations. In this context, we review prior work to highlight this gap and introduce our targeted approach.

2.1 Label Flipping and Data Poisoning Attacks

Data poisoning attacks aim to corrupt the training process of a model by injecting malicious data [5, 10, 41]. These attacks can be broadly categorized by their goal, such as degrading overall model performance (availability attacks) or causing specific misclassifications (integrity attacks or backdoors) [12, 27, 40]. Our work

focuses on a classic and practical form of availability attack: label flipping [23, 26, 31].

Early work on label flipping explored its impact on traditional classifiers like SVMs and Logistic Regression [4, 24]. With the rise of deep learning, research has demonstrated the vulnerability of deep neural networks to even small fractions of flipped labels [29, 37]. Attack strategies have evolved from simple random flipping to more sophisticated, targeted approaches that select specific samples or class pairs to maximize damage [16, 34]. However, a common thread in these studies is the focus on designing more effective attacks, often assuming uniform vulnerability or selecting targets based on simple heuristics. In contrast, our work systematically investigates why these attacks have a differential impact, providing a theoretical foundation for understanding and predicting this unequal vulnerability.

2.2 Model Vulnerability and Class Separability

Understanding the intrinsic vulnerabilities of deep learning models is a critical area of research. Much work has focused on test-time vulnerabilities, particularly adversarial examples [11, 28]. A parallel line of work has explored how class-level properties, such as inter-class similarity or feature space geometry, affect model performance and robustness [1, 25]. For instance, it has been shown that class imbalance can lead to unequal robustness against adversarial attacks [33].

Our work contributes to this area by establishing a direct, theoretical link between a specific geometric property and a model’s vulnerability to training-time label flipping attacks. While the notion that similar classes are harder to learn is intuitive, we are the first to formalize this relationship in the context of poisoning vulnerability and use it to build a principled, targeted defense. Our analysis of the differential impact of attacks provides a new, fine-grained perspective on model robustness that goes beyond average performance metrics.

3 Theoretical Analysis of Unequal Vulnerability

In this section, we develop a theoretical framework to formally explain why the vulnerability to label flipping attacks is unequal across different class pairs. We construct a simplified, yet insightful, generative model where class confusability can be precisely defined and its impact on attack efficacy can be analytically derived. Our analysis reveals that the standardized separation between classes is the fundamental factor governing this differential vulnerability.

3.1 Problem Formulation and Definitions

To make the analysis tractable, we consider a d -dimensional binary classification problem. Data points $(\mathbf{x}, y)$ are drawn from a distribution D , where the class label $y \in \{+1, -1\}$. The class-conditional distributions are modeled as multivariate Gaussians with isotropic covariance:

- Class +1: $p(\mathbf{x}|y = +1) = \mathcal{N}(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$
- Class -1: $p(\mathbf{x}|y = -1) = \mathcal{N}(-\boldsymbol{\mu}, \sigma^2 \mathbf{I})$

where $\boldsymbol{\mu} = (\eta, 0, \dots, 0)^\top$ with $\eta > 0$, and $\mathbf{I}$ is the d -dimensional identity matrix. We assume equal prior probabilities, $P(y = +1) = P(y = -1) = 0.5$. In this model, the difficulty of distinguishing

between the two classes is controlled by the relationship between their separation distance and their variance.

We now introduce the core definitions for our analysis.

DEFINITION 1 (STANDARDIZED SEPARATION). *The Standardized Separation, $S(\eta, \sigma)$, between the two classes is defined as the 1/2 Euclidean distance between their centroids, normalized by the standard deviation:*

$$S(\eta, \sigma) = \frac{\|\mu - (-\mu)\|_2}{2\sigma} = \frac{\eta}{\sigma}. \quad (1)$$

A smaller $S(\eta, \sigma)$ implies higher intrinsic class confusability.

DEFINITION 2 (ATTACK IMPACT). *The impact of a label flipping attack, Impact, is defined as the increase in the misclassification rate on the clean test data of the source class (here, Class -1) when the model is trained on a poisoned dataset versus a clean one:*

$$\text{Impact} = \text{Err}_{\text{poisoned}}(y = -1) - \text{Err}_{\text{clean}}(y = -1). \quad (2)$$

Our goal is to show that Impact is a monotonically decreasing function of the Standardized Separation $S(\eta, \sigma)$.

3.2 Optimal Classifier on Clean and Poisoned Data

We first establish the decision boundary for an optimal linear classifier under both clean and poisoned training scenarios. We focus on linear classifiers of the form $f(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x} + b)$.

LEMMA 1. *For the clean data distribution D , the Bayes-optimal linear decision boundary that minimizes the classification error is the hyperplane $x_1 = 0$.*

PROOF SKETCH. It is well-established that for two Gaussian distributions with equal covariance and priors, the optimal decision boundary is the perpendicular bisector of the segment connecting their means. Given μ and $-\mu$, this bisector is the hyperplane $x_1 = 0$.

Next, we analyze how this boundary shifts under a label flipping attack where a fraction p of the training samples from Class -1 have their labels flipped to +1.

LEMMA 2. *For a linear classifier trained on the dataset poisoned with a label-flipping ratio p , the optimal decision boundary shifts from $x_1 = 0$ to $x_1 = -\delta$, where $\delta = \eta \frac{p}{1+p}$.*

PROOF SKETCH. After the attack, the empirical mean of the data labeled as +1 shifts to $\mu_p = \frac{1-p}{1+p}\mu$. The optimal linear separating hyperplane, approximated by the perpendicular bisector between the new mean μ_p and the original mean $-\mu$ of Class -1, is located at $x_1 = -\eta \frac{p}{1+p}$. Thus, the boundary shifts by $\delta > 0$.

Figure 2 provides a visual illustration of these concepts and foreshadows our main theorem: a larger Standardized Separation S mitigates the impact of the boundary shift δ .

3.3 Main Theorem: Vulnerability is Governed by Separation

With the decision boundaries established, we can now state and prove our main theorem.

THEOREM 1. *The impact of a label flipping attack is a monotonically decreasing function of the Standardized Separation $S(\eta, \sigma)$.*

PROOF. First, we compute the error rate for the source class (-1) on clean test data. The first component of a sample from Class -1, x_1 , follows $\mathcal{N}(-\eta, \sigma^2)$.

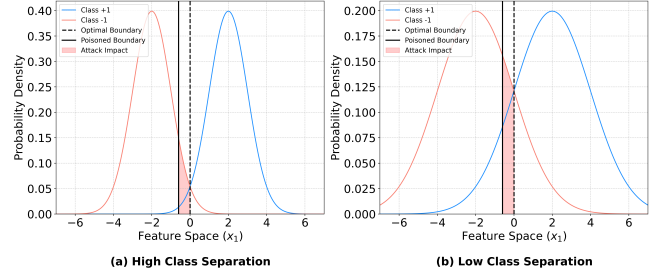


Figure 2: Illustration of attack impact under different Standardized Separation ($S = \eta/\sigma$) values. The solid curves represent the probability density of Class +1 (centered at η) and Class -1 (centered at $-\eta$). The dashed line is the optimal boundary on clean data ($x_1 = 0$), while the solid line is the boundary after a poisoning attack ($x_1 = -\delta$). The shaded area represents the attack impact—the additional error on Class -1. (a) With a high S , the classes are well-separated. The boundary shift results in a small impact. (b) With a low S , the classes are highly confusable. The same attack causes a much larger impact.

The error of the clean classifier (boundary at $x_1 = 0$) is:

$$\text{Err}_{\text{clean}} = P(x_1 > 0) = \Phi\left(\frac{0 - \eta}{\sigma}\right) = \Phi\left(-\frac{\eta}{\sigma}\right), \quad (3)$$

where $\Phi(\cdot)$ is the CDF of the standard normal distribution.

The error of the poisoned classifier (boundary at $x_1 = -\delta$), using Lemma 2, is:

$$\begin{aligned} \text{Err}_{\text{poisoned}} &= P(x_1 > -\delta) \\ &= \Phi\left(-\frac{\eta}{\sigma} \left(1 + \frac{\delta}{\eta}\right)\right) \\ &= \Phi\left(-\frac{\eta}{\sigma} \left(1 + \frac{p}{1+p}\right)\right) = \Phi\left(-\frac{\eta}{\sigma} \frac{1+2p}{1+p}\right). \end{aligned} \quad (4)$$

The attack impact from Definition 2 is therefore:

$$\text{Impact} = \Phi\left(-\frac{\eta}{\sigma} \frac{1+2p}{1+p}\right) - \Phi\left(-\frac{\eta}{\sigma}\right). \quad (5)$$

Let $S = S(\eta, \sigma) = \eta/\sigma$ be the Standardized Separation (from Eq. 1) and let $C_p = \frac{1+2p}{1+p} > 1$ be a constant dependent only on the poison ratio. The impact can be written as a function of S :

$$\text{Impact}(S) = \Phi(-SC_p) - \Phi(-S). \quad (6)$$

To prove that $\text{Impact}(S)$ is monotonically decreasing, we compute its derivative with respect to S :

$$\frac{d(\text{Impact})}{dS} = \phi(-SC_p) (-C_p) - \phi(-S) (-1), \quad (7)$$

where $\phi(\cdot)$ is the PDF of the standard normal distribution. This simplifies to:

$$\frac{d(\text{Impact})}{dS} = [\phi(S) - C_p \phi(SC_p)], \quad (8)$$

where we used the property $\phi(-z) = \phi(z)$. For $S > 0$ and $C_p > 1$, this derivative is negative. Intuitively, as S increases, the exponential decay of the Gaussian PDF $\phi(\cdot)$ causes the term $C_p \phi(SC_p)$ to diminish much faster than $\phi(S)$, ensuring the bracket becomes negative. Thus, a larger Standardized Separation S leads to a smaller attack impact.

This theorem provides the theoretical foundation for our work, directly linking the empirically observed unequal vulnerability to the fundamental geometric property of class separability.

4 Confusability-Aware Contrastive Learning (CACL)

Our theoretical analysis establishes a direct link between a class pair's vulnerability and its Standardized Separation. This finding calls for a defense mechanism that intelligently allocates its resources to fortify the most vulnerable parts of the feature space. Instead of applying a uniform regularization across all classes (a strategy that may be inefficient or even detrimental), we propose a targeted approach that is explicitly driven by our vulnerability metric.

We introduce Confusability-Aware Contrastive Learning (CACL), a novel regularization framework designed to proactively enhance the separability of classes based on their intrinsic confusability. The core principle of CACL is to modulate the strength of a contrastive learning objective with a weight derived directly from our empirical confusability index.

4.1 Vulnerability-Aware Weighting Scheme

The CACL framework is guided by a quantitative measure of each class pair's vulnerability. We compute an empirical confusability index, $C(i, j)$, for each pair (i, j) based on their feature-space statistics from a pre-trained probe model.

$$C(i, j) = \frac{\text{var}_i + \text{var}_j}{\|\mathbf{c}_i - \mathbf{c}_j\|_2^2}. \quad (9)$$

Instead of using this index to make a hard selection, we use it to create a continuous weighting scheme. For each class pair (i, j) , we define a weight $w(i, j)$ that is a monotonic function of their confusability. A simple and effective choice is a normalized weight derived via the softmax function:

$$w(i, j) = \frac{\exp(C(i, j)/T)}{\sum_{i' \neq j'} \exp(C(i', j')/T)}, \quad (10)$$

where the sum is over all distinct class pairs, and T is a temperature parameter that controls the sharpness of the weighting distribution. A lower T concentrates the weights on the most confusable pairs. This weighting scheme allows our defense to be both comprehensive and focused.

4.2 The CACL Objective Function

The CACL method integrates this weighting scheme into a prototypical contrastive loss framework. The total training loss is a

weighted sum of the standard Cross-Entropy loss and the CACL regularization term:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{CACL}}, \quad (11)$$

where λ is a balancing hyperparameter.

The $\mathcal{L}_{\text{CACL}}$ term is an aggregation of pairwise contrastive losses, where each pair's contribution is scaled by its confusability weight. For a sample $\mathbf{z}_i$ from class y_i with its class centroid $\mathbf{c}_{y_i}$, the loss is based on comparing its affinity to its own class prototype versus other class prototypes.

Let $\text{sim}(\cdot, \cdot)$ be the cosine similarity and τ be the temperature for the contrastive loss. To simplify the notation, we define a scoring function $s(\mathbf{z}, \mathbf{c})$ that measures the affinity between a feature vector $\mathbf{z}$ and a class centroid $\mathbf{c}$ as:

$$s(\mathbf{z}, \mathbf{c}) = \exp(\text{sim}(\mathbf{z}, \mathbf{c})/\tau). \quad (12)$$

Using this scoring function, the CACL loss component for a single sample $\mathbf{z}_i$ from class y_i is then expressed as:

$$\mathcal{L}_{\text{CACL}}(\mathbf{z}_i) = -\log \frac{s(\mathbf{z}_i, \mathbf{c}_{y_i})}{s(\mathbf{z}_i, \mathbf{c}_{y_i}) + \sum_{j \neq y_i} w(y_i, j) \cdot s(\mathbf{z}_i, \mathbf{c}_j)}, \quad (13)$$

where $w(y_i, j)$ is the pre-computed confusability weight for the pair (y_i, j) .

The total regularization term for a mini-batch $\mathcal{B}$ is the average of these individual losses:

$$\mathcal{L}_{\text{CACL}} = \frac{1}{|\mathcal{B}|} \sum_{\mathbf{z}_i \in \mathcal{B}} \mathcal{L}_{\text{CACL}}(\mathbf{z}_i). \quad (14)$$

By incorporating the confusability weights $w(i, j)$ directly into the loss, CACL forces the model to expend more effort on separating class pairs that are intrinsically hard to distinguish (high $C(i, j)$), while applying a gentler, non-zero regularization to less confusable pairs. This soft, targeted approach directly operationalizes our theoretical findings in a fully differentiable manner. The implementation details of the weighting computation and training loop are provided in Appendix A (Algorithms 1 and 2).

5 Experiments

In this section, we conduct a comprehensive set of experiments to empirically validate our theoretical framework and evaluate the effectiveness of our proposed defense method, CACL. Our evaluation is designed to answer two key questions:

- (1) Does our theoretical model correctly predict the attack vulnerability on real-world datasets?
- (2) How effective is CACL in mitigating label flipping attacks, especially when compared to standard training and non-targeted robust methods?

5.1 Experimental Setup

Datasets. Our primary analysis is conducted on **CIFAR-10** [19]. To evaluate the broader applicability of our findings, we also perform experiments on **CIFAR 100** [19], which presents a more fine-grained classification challenge, and **Fashion-MNIST** [30], a simpler grayscale image dataset.

Models. For fair comparison, we use the widely-adopted **ResNet-18** [13] architecture as our primary model for all experiments. Besides, we also conduct comparative analyses using a simple **Multi-Layer Perceptron (MLP)** and a modern **Vision Transformer (ViT)** [7] architecture.

Attack Protocol. We simulate a label flipping attack where an adversary poisons the training set by flipping the labels of a fraction p of samples from a chosen source class to a target class. We set the poison ratio $p = 10\%$ for most experiments. To further understand the relationship between attack intensity and impact, supplementary experiments are run with varying poison ratios.

Baselines. We evaluate our proposed Confusability-Aware Contrastive Learning (CACL) against a selection of strong baselines representing different approaches to robust learning:

- **Standard CE:** A model trained with the standard Cross-Entropy loss, serving as a non-robust baseline.
- **SupCon:** Supervised Contrastive learning [18], a powerful framework for learning discriminative representations by contrasting samples in the feature space.
- **TCL:** Twin Contrastive Learning [14], a recent, high-performing method for learning from noisy labels.
- **SGN:** Robust Classification via Regression [9], a novel approach that recasts noisy-label classification as a regression problem.

Implementation Details. All models are trained for 100 epochs with a batch size of 128. The data is preprocessed with standard normalization without any data augmentation. We use an SGD optimizer with an initial learning rate of 0.1, a momentum of 0.9, and a weight decay of $5e-4$. For CACL, the hyperparameters are set to $\lambda = 0.5$, $\tau = 0.1$, and $T = 1.0$, chosen based on a held-out validation set. The confusability indices used for the CACL weighting scheme are determined from the vulnerability assessment stage. All experiments are repeated over three random seeds, and we report the average results. Detailed information on model architectures and a full list of hyperparameters can be found in Appendix B.

5.2 Simulation Study: Validating the Theory

To provide a clean and controlled validation of our theoretical claims in Theorem 1, we first conduct a simulation study on 2D synthetic Gaussian data. We generate data for two classes according to our theoretical model $p(\mathbf{x}|y = \pm 1) = \mathcal{N}(\pm\boldsymbol{\mu}, \sigma^2\mathbf{I})$, where $\boldsymbol{\mu} = (\eta, 0)^\top$. We then perform a label flipping attack with $p = 10\%$ and measure the resulting accuracy drop on the source class.

As shown in Figure 3, the experimental results on synthetic data perfectly align with our theoretical predictions. Figure 3(a) shows that for a fixed noise level σ , the attack impact monotonically decreases as the class separation η increases. Conversely, Figure 3(b) shows that for a fixed separation η , the impact increases with the noise level σ . Most importantly, Figure 3(c) and (d) demonstrate that these two factors are unified by the Standardized Separation $S = \eta/\sigma$. Figure 3(c) shows a clear monotonic decreasing relationship between the attack impact and S .

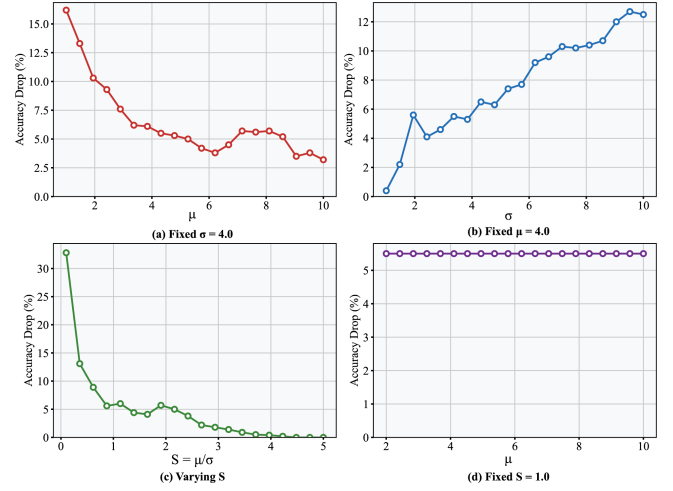


Figure 3: Simulation results validating our theoretical framework. (a) Attack impact decreases as separation η increases (with fixed σ). (b) Attack impact increases as noise σ increases (with fixed η). (c, d) Attack impact is significantly correlated with the Standardized Separation $S = \eta/\sigma$.

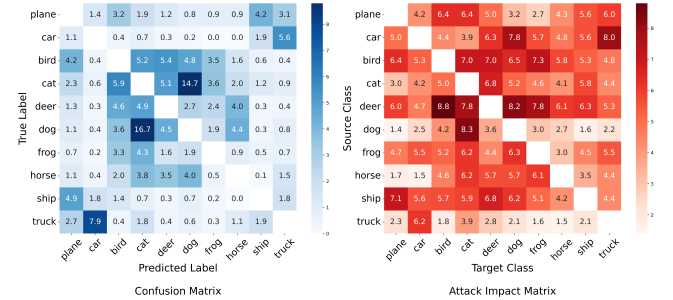


Figure 4: The strong positive correlation between Baseline Confusion Rate (a proxy for class separability) and the Attack Impact (test accuracy drop on the source class) on CIFAR-10. Each point represents a unique source-target class pair for a label flipping attack ($p = 10\%$).

5.3 Vulnerability Analysis on Real-World Datasets

5.3.1 Main Results. Having validated our theory on a simplified model, we now investigate whether the same principles hold for deep neural networks on real-world data. Our theory posits that the fundamental determinant of vulnerability is the standardized separation between classes. For complex models, this geometric property is not directly accessible. However, it is reasonable to assume that a lower Standardized Separation in the latent feature space will manifest as a higher confusion rate at the model’s output layer. Therefore, we use the baseline confusion matrix of a clean model as a practical and powerful proxy to measure the intrinsic class separability. Our goal is to empirically test the correlation between this proxy and the actual measured attack impact on the source class.

The experimental procedure and its results are presented in Figure 4. To obtain our vulnerability predictor, we first train a standard ResNet-18 on clean data and compute its baseline confusion matrix, M , where an entry M_{ij} represents the misclassification rate from source class i to target class j . Subsequently, we conduct 90 independent label flipping attacks, one for each distinct class pair (i, j) , with a poisoning rate of 10%. For each attack, we measure the attack impact as the resulting drop in test accuracy on the source class i .

As hypothesized, the results reveal a remarkably strong positive correlation between the baseline confusion rate (M_{ij}) and the measured attack impact on the source class, with a high Pearson correlation coefficient of $\rho = 0.77$. This provides powerful empirical evidence that the abstract concept of Standardized Separation, manifested through the observable confusion matrix, is a key predictor of vulnerability. In essence, the more a model already struggles to distinguish a source class from a target, the more its ability to recognize the source class will degrade when poisoned towards that target.

Table 1: Pearson (ρ_p) and Spearman (ρ_s) correlation coefficients between baseline confusion and attack impact across all experimental settings. The consistently high values confirm the generality of our hypothesis.

Model	Dataset	Pearson Corr. (ρ_p)	Spearman Corr. (ρ_s)
MLP	Fashion-MNIST	0.81	0.83
	CIFAR-10	0.74	0.76
	CIFAR-100 (subset)	0.71	0.75
ResNet-18	Fashion-MNIST	0.85	0.86
	CIFAR-10	0.75	0.79
	CIFAR-100 (subset)	0.74	0.78
ViT	Fashion-MNIST	0.83	0.84
	CIFAR-10	0.76	0.78
	CIFAR-100 (subset)	0.73	0.76

5.3.2 Vulnerability Across Diverse Models and Datasets. To ascertain that the observed link between baseline class confusability and attack vulnerability is a fundamental principle rather than an artifact of a specific architecture or dataset, we conducted a broad generalizability study. We tested our core hypothesis across three distinct model paradigms: a simple Multi-Layer Perceptron (MLP), a modern convolutional network (ResNet-18), and a Vision Transformer (ViT). These models were evaluated on three datasets with varying complexity: Fashion-MNIST (simple, grayscale), CIFAR-10 (moderate complexity), and CIFAR-100 (fine-grained). For CIFAR-100, to maintain visualization clarity, we selected 10 representative classes and performed pairwise attacks among them.

The results are presented in Figure 5. Each subplot corresponds to a model architecture, plotting the attack impact against the baseline confusion rate for all three datasets. A clear and consistent positive correlation emerges across every experimental setting. Regardless of whether the model relies on simple linear layers, residual connections, or self-attention mechanisms, it is systematically more vulnerable to label flipping attacks between classes it already finds difficult to distinguish.

Table 1 provides a quantitative summary of these findings, reporting both Pearson (ρ_p) and Spearman (ρ_s) correlation coefficients. The consistently high values for both metrics confirm a strong, monotonic relationship.

5.3.3 Vulnerability under Different Poisoning Rates. To further explore the nature of this unequal vulnerability, we investigated its behavior under varying levels of attack intensity. We conducted experiments on CIFAR-10 with ResNet-18, systematically varying the poison ratio p across a range of {10%, 20%, 30%, 40%}. For each rate, we performed targeted attacks from a single source class ('dog') to all other nine classes.

The results, shown in Figure 6, demonstrate a robust correlation between confusability and vulnerability across all tested attack intensities. For each poison rate, the pattern of accuracy drops among target classes closely follows the pattern of the baseline confusion rates. This confirms that the predictive power of class confusability is stable, with the attack intensity primarily modulating the severity of the damage rather than the underlying vulnerability pattern.

5.4 Evaluation of Defense Method

After determining the predictable nature of vulnerability, we now evaluate the effectiveness of our proposed CACL framework against label flipping attacks. We test all methods under two scenarios: a **Targeted Attack**, which concentrates the poison budget on the most confusable class pairs to simulate a worst-case scenario, and a **Random Attack**, which distributes the poison across randomly chosen pairs.

5.4.1 Main Results. The main results of our defense evaluation on CIFAR-10 are summarized in Table 2. The data shows that state-of-the-art methods like SGN establish a highly competitive baseline, achieving top-tier performance on clean data and in overall accuracy under a random attack. In the worst-case targeted attack, however, our CACL method proves to be significantly more effective. It achieves the highest overall accuracy among all methods. Most compellingly, CACL elevates the worst-class accuracy to 73.50%, substantially outperforming all baselines, with the next best method, SGN, at 68.80%. This confirms that our targeted strategy of selectively increasing the separation for the most confusable pairs is uniquely effective at preventing catastrophic failure on the most vulnerable classes.

Furthermore, CACL's ability to foster robustness equity is demonstrated under the general random attack. While its overall accuracy is highly competitive, its key achievement is yielding the lowest Accuracy Standard Deviation (6.2) by a significant margin. This indicates that CACL's protection is far more evenly distributed across all classes, directly addressing the core problem of unequal vulnerability. This demonstrates that CACL provides a fairer and more uniformly resilient defense against structured data poisoning. These advantages are consistently observed in our additional experiments on CIFAR-100 and Fashion-MNIST, which are detailed in the Appendix.

5.4.2 Feature Space Visualization. To visualize how CACL reshapes the feature representation, we use t-SNE [22] to project the feature vectors from the test set into a 2D space. We focus on a particularly vulnerable pair identified in our analysis: 'cat' and 'dog'. Figure 7

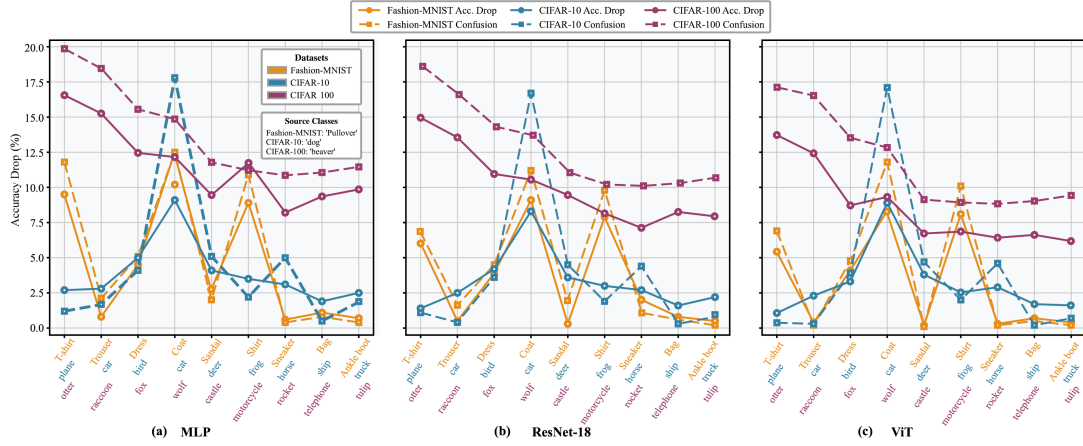


Figure 5: The confusability-vulnerability link across diverse models and datasets. Each point represents a source-target attack pair ($p = 10\%$). Each subplot compares the attack impact (accuracy drop, solid lines) against the baseline confusion rate (dashed lines) for a fixed source class attacked towards various target classes. The experiment is repeated across three model architectures (sub-plots) and three datasets (colors).

Table 2: Defense method performance on CIFAR-10 under a 10% poison rate. The Targeted Attack evaluates a worst-case scenario on the ‘dog’ $\rightarrow$ ‘cat’ pair. The Random Attack evaluates general robustness across multiple random pairs. “Worst-Class Acc” is the accuracy of the ‘dog’ class post-attack. “Acc. Std. Dev” measures performance variance across all 10 classes. Bold indicates the best performance in each column.

Method	Clean Acc. (%)	Targeted Attack (Worst-Case)		Random Attack		
		Overall Acc. (%)	Worst-Class Acc. (%)	Overall Acc. (%)	Acc. Drop $\downarrow$	Acc. Std. Dev. $\downarrow$
Standard CE	79.53	74.36	65.78	75.13	4.40	9.8
SupCon	79.98	75.88	68.43	77.15	2.83	8.5
TCL	79.62	75.95	68.10	77.81	1.81	7.4
SGN	80.11	76.10	68.80	77.85	2.26	7.9
CACL (Ours)	79.95	77.52	73.50	77.65	2.30	6.2

compares the feature distributions of models trained with Standard CE and our CACL under a targeted ‘cat’ $\rightarrow$ ‘dog’ attack.

As seen in Figure 7(a), the attack causes the feature clusters of ‘cat’ and ‘dog’ to **severely overlap**, making them largely indistinguishable. In contrast, Figure 7(b) shows that the model trained with CACL is significantly more resilient, with the degree of separation drastically improved and the majority of samples forming distinct clusters. This provides direct evidence that CACL successfully fortifies the feature space by pushing confusable classes apart, mitigating the attack at its root.

5.4.3 Impact of Weighting Scheme. We compare our main method, **CACL (Weighted)**, with its simpler variant, **CACL (Hard-Targeting)**, which applies a uniform loss only to the Top-K vulnerable pairs. As shown in Table 3, the soft-weighting scheme of CACL consistently outperforms the hard-targeting variant, especially in preserving clean accuracy. This suggests that the nuanced, continuous weighting is more effective than a binary selection, likely by avoiding overly aggressive regularization on moderately confusable pairs.

Table 3: Ablation study on the weighting scheme. Weighted CACL provides a better trade-off between clean accuracy and robustness.

Method Variant	Clean Acc. (%)	Worst-Class Acc. (%)
CACL (Hard-Targeting)	78.28	71.65
CACL (Weighted)	79.95	73.50

5.5 The Indirect Effect

An intriguing finding from our experiments is that label flipping attacks can induce non-trivial **collateral impacts** on classes that are not directly involved in the attack. Figure 8 provides a compelling example of this phenomenon. We conduct a ‘ship’ $\rightarrow$ ‘car’ attack and observe its effect on the ‘truck’ class, which is semantically close to both. The confusion matrices in Figure 8 show that after the attack, the number of ‘truck’ samples misclassified as ‘car’ drops from 79 to 60. This, in turn, leads to an unexpected *improvement* in the classification accuracy of the ‘truck’ class, from 82.9% to 84.9%.

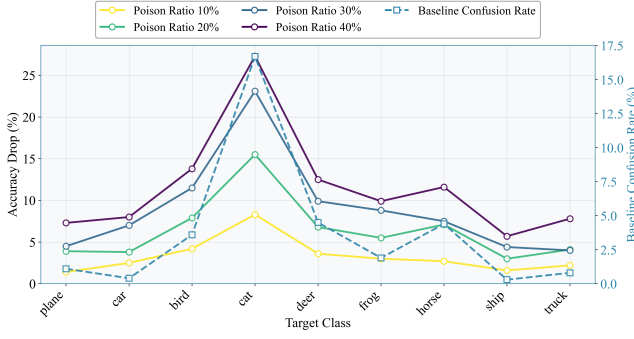


Figure 6: The Vulnerability Pattern Persists Across Different Poisoning Rates. The plot shows the attack impact (accuracy drop, colored solid lines) on the ‘dog’ source class for various target classes on CIFAR-10 (ResNet-18). Each colored line represents a different poison rate (p). The dashed line shows the constant baseline confusion rate for comparison.

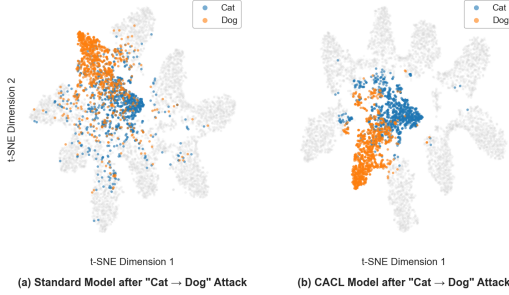


Figure 7: t-SNE visualization of feature representations for the ‘cat’ (blue) and ‘dog’ (orange) classes from the CIFAR-10 test set. (a) Under a targeted ‘cat’→‘dog’ attack, the standard model’s feature clusters collapse. (b) The CACL-trained model exhibits significantly reduced overlap and better cluster separation, demonstrating its robustness.

We hypothesize that this is due to a feature-space repulsion effect. To accommodate the poisoned ‘ship’ samples, the model is forced to adjust its decision boundary for ‘car’, effectively pushing the ‘car’ cluster’s boundary away from the ‘ship’ cluster. As the ‘truck’ cluster is also in the vicinity, this boundary adjustment inadvertently increases the separation between the ‘car’ and ‘truck’ clusters as well. This collateral benefit highlights the complex, non-local nature of how model representations adapt to poisoning. It underscores that an attack on one class pair can trigger a cascade of geometric shifts across the entire feature space, leading to unexpected outcomes for bystander classes.

6 Discussion: Beyond Poisoning

While this paper has focused on the threat of malicious label flipping attacks, our findings offer a more general perspective on model robustness. A label flipping attack from a source class i to a target class j can be fundamentally reinterpreted as the injection of a specific, structured form of class-to-class label noise. Consequently,

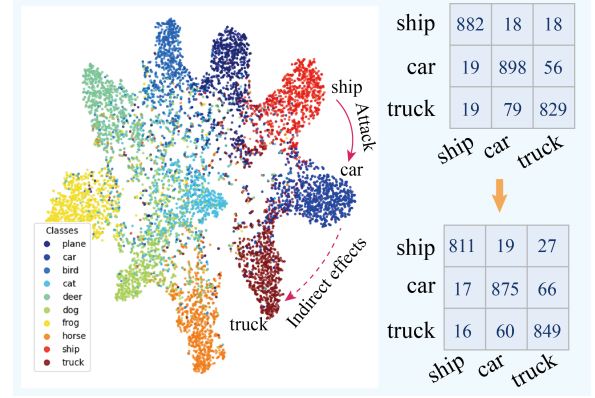


Figure 8: Indirect effects of a ‘ship’→‘car’ attack. The t-SNE plot shows the feature space of related classes. The confusion matrices (right) show results for ‘ship’, ‘car’, and ‘truck’ before (top) and after (bottom) the attack.

our entire framework can be viewed not just as a defense against a particular adversary, but as a tool for analyzing and enhancing a model’s fine-grained noise robustness.

Our theoretical analysis, which links vulnerability to the standardized separation $S(i, j)$, is not inherently adversarial. It simply quantifies how the geometry of two class distributions makes one susceptible to being confused for the other, and is thus directly applicable to understanding the impact of any process that causes class i samples to be labeled as class j .

From this perspective, CACL transcends its role as a mere defense mechanism to become a principled technique for improving robustness against structured label noise. By identifying class pairs with the lowest separability and selectively increasing their feature-space distance, it effectively de-noises the representation space at its most vulnerable points. This represents a step towards building models that are specifically hardened against plausible forms of real-world label corruption.

7 Conclusion

In this paper, we investigated the unequal vulnerability of machine learning models to label flipping attacks. We demonstrated, both theoretically and empirically, that the attack’s impact is fundamentally governed by the standardized separation between classes—lower separability leads to significantly higher vulnerability. Grounded in this insight, we proposed Confusability-Aware Contrastive Learning (CACL), a targeted defense that proactively fortifies the most confusable class pairs by maximizing their feature-space distance. Extensive experiments confirmed that CACL provides superior protection compared to standard and non-targeted robust training methods, especially for the most vulnerable classes. Our work provides a new, structural perspective on data poisoning and offers a principled approach to building more equitably robust models.

Acknowledgments

This work is supported by the the Natural Science Foundation of China (Grant No. NSFC 42550187 and 62476191).

References

- [1] Uduak Idio Akpan and Andrew Starkey. 2021. Review of classification algorithms with changing inter-class distances. *Machine Learning with Applications* 4 (2021), 100031.
- [2] Junaid Akram, Ali Anaissi, Awais Akram, Rajkumar Singh Rathore, and Rutvij H Jhaveri. 2024. Adversarial label-flipping attack and defense for anomaly detection in spatial crowdsourcing uav services. *IEEE Transactions on Consumer Electronics* (2024).
- [3] Sarvagya Bhargava and Mark Stamp. 2025. Robustness of Selected Learning Models Under Label-Flipping Attack. In *Machine Learning, Deep Learning and AI for Cybersecurity*. Springer, 493–506.
- [4] Battista Biggio, Blaine Nelson, and Pavel Laskov. 2011. Support vector machines under adversarial label noise. In *Asian conference on machine learning*. PMLR, 97–112.
- [5] Battista Biggio, Blaine Nelson, and Pavel Laskov. 2012. Poisoning attacks against support vector machines. *arXiv preprint arXiv:1206.6389* (2012).
- [6] Antonio Emanuele Cinà, Kathrin Grosse, Ambra Demontis, Sebastiano Vascon, Werner Zellinger, Bernhard A Moser, Alina Oprea, Battista Biggio, Marcello Pelillo, and Fabio Roli. 2023. Wild patterns reloaded: A survey of machine learning security against training data poisoning. *Comput. Surveys* 55, 13s (2023), 1–39.
- [7] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).
- [8] Abdessamad El-Kabid and El-Mahdi El-Mhamdi. 2025. Approaching the Harm of Gradient Attacks While Only Flipping Labels. *arXiv preprint arXiv:2503.00140* (2025).
- [9] Erik Englesson and Hossein Azizpour. 2024. Robust classification via regression for learning with noisy labels. In *ICLR 2024-The Twelfth International Conference on Learning Representations, Messe Wien Exhibition and Congress Center, Vienna, Austria, May 7-11t, 2024*.
- [10] Micah Goldblum, Dimitris Tsipras, Chulin Xie, Xinyun Chen, Avi Schwarzschild, Dawn Song, Aleksander Mądry, Bo Li, and Tom Goldstein. 2022. Dataset security for machine learning: Data poisoning, backdoor attacks, and defenses. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 2 (2022), 1563–1580.
- [11] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. 2014. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572* (2014).
- [12] Tianyu Gu, Brendan Dolan-Gavitt, and Siddharth Garg. 2017. Badnets: Identifying vulnerabilities in the machine learning model supply chain. *arXiv preprint arXiv:1708.06733* (2017).
- [13] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778.
- [14] Zhizhong Huang, Junping Zhang, and Hongming Shan. 2023. Twin contrastive learning with noisy labels. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 11661–11670.
- [15] Shahana Ibrahim, Panagiotis A Traganitis, Xiao Fu, and Georgios B Giannakis. 2025. Learning from crowdsourced noisy labels: A signal processing perspective. *IEEE Signal Processing Magazine* 42, 3 (2025), 84–106.
- [16] Rishi Jha, Jonathan Hayase, and Sewoong Oh. 2023. Label poisoning is all you need. *Advances in Neural Information Processing Systems* 36 (2023), 71029–71052.
- [17] Yifeng Jiang, Weiwen Zhang, and Yanxi Chen. 2023. Data quality detection mechanism against label flipping attacks in federated learning. *IEEE Transactions on Information Forensics and Security* 18 (2023), 1625–1637.
- [18] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. 2020. Supervised contrastive learning. *Advances in neural information processing systems* 33 (2020), 18661–18673.
- [19] Alex Krizhevsky, Geoffrey Hinton, et al. 2009. Learning multiple layers of features from tiny images. (2009).
- [20] Li Li, Yuxi Fan, Mike Tse, and Kuo-Yi Lin. 2020. A review of applications in federated learning. *Computers & Industrial Engineering* 149 (2020), 106854.
- [21] Yiming Li, Yong Jiang, Zhifeng Li, and Shu-Tao Xia. 2022. Backdoor learning: A survey. *IEEE transactions on neural networks and learning systems* 35, 1 (2022), 5–22.
- [22] Laurens van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-SNE. *Journal of machine learning research* 9, Nov (2008), 2579–2605.
- [23] Orson Mengara. 2024. A backdoor approach with inverted labels using dirty label-flipping attacks. *IEEE Access* (2024).
- [24] Blaine Nelson, Marco Barreno, Fuching Jack Chi, Anthony D Joseph, Benjamin IP Rubinstein, Udam Saini, Charles Sutton, J Doug Tygar, and Kai Xia. 2008. Exploiting machine learning to subvert your spam filter. *Leet* 8, 1 (2008), 9.
- [25] Vardan Papayan, XY Han, and David L Donoho. 2020. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences* 117, 40 (2020), 24652–24663.
- [26] Elan Rosenfeld, Ezra Winston, Pradeep Ravikumar, and Zico Kolter. 2020. Certified robustness to label-flipping attacks via randomized smoothing. In *International Conference on Machine Learning*. PMLR, 8230–8241.
- [27] Ali Shafahi, W Ronny Huang, Mahyar Najibi, Octavian Suciu, Christoph Studer, Tudor Dumitras, and Tom Goldstein. 2018. Poison frogs! targeted clean-label poisoning attacks on neural networks. *Advances in neural information processing systems* 31 (2018).
- [28] Christian Szegedy, Wojciech Zaremba, Ilya Sutskever, Joan Bruna, Dumitru Erhan, Ian Goodfellow, and Rob Fergus. 2013. Intriguing properties of neural networks. *arXiv preprint arXiv:1312.6199* (2013).
- [29] Huang Xiao, Battista Biggio, Gavin Brown, Giorgio Fumera, Claudia Eckert, and Fabio Roli. 2015. Is feature selection secure against training data poisoning?. In *international conference on machine learning*. PMLR, 1689–1698.
- [30] Han Xiao, Kashif Rasul, and Roland Vollgraf. 2017. Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747* (2017).
- [31] Han Xiao, Huang Xiao, and Claudia Eckert. 2012. Adversarial label flips attack on support vector machines. In *ECAI 2012*. IOS Press, 870–875.
- [32] Junlin Xie, Zhihong Chen, Ruifei Zhang, Xiang Wan, and Guanbin Li. 2024. Large multimodal agents: A survey. *arXiv preprint arXiv:2402.15116* (2024).
- [33] Han Xu, Xiaorui Liu, Yaxin Li, Anil Jain, and Jiliang Tang. 2021. To be robust or to be fair: Towards fairness in adversarial training. In *International conference on machine learning*. PMLR, 11492–11501.
- [34] Qianqian Xu, Zhiyong Yang, Yunrui Zhao, Xiaochun Cao, and Qingming Huang. 2022. Rethinking label flipping attack: From sample masking to sample thresholding. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 6 (2022), 7668–7685.
- [35] Shanqing Yu, Jie Shen, Shaocong Xu, Jinhuan Wang, Zeyu Wang, and Qi Xuan. 2025. Label-Flipping Attacks in GNN-Based Federated Learning. *IEEE Transactions on Network Science and Engineering* (2025).
- [36] Shanhao Zhan, Lianfen Huang, Gaoyu Luo, Shaolong Zheng, Zhibin Gao, and Han-Chieh Chao. 2025. A Review on Federated Learning Architectures for Privacy-Preserving AI: Lightweight and Secure Cloud-Edge-End Collaboration. *Electronics* 14, 13 (2025), 2512.
- [37] Hongpo Zhang, Ning Cheng, Yang Zhang, and Zhanbo Li. 2021. Label flipping attacks against Naive Bayes on spam filtering systems. *Applied Intelligence* 51, 7 (2021), 4503–4514.
- [38] Jing Zhang. 2022. Knowledge learning with crowdsourcing: A brief review and systematic perspective. *IEEE/CAA Journal of Automatica Sinica* 9, 5 (2022), 749–762.
- [39] Mengmei Zhang, Linmei Hu, Chuan Shi, and Xiao Wang. 2020. Adversarial label-flipping attack and defense for graph neural networks. In *2020 IEEE International Conference on Data Mining (ICDM)*. IEEE, 791–800.
- [40] Shaobo Zhang, Yimeng Pan, Qin Liu, Zheng Yan, Kim-Kwang Raymond Choo, and Guojun Wang. 2024. Backdoor attacks and defenses targeting multi-domain ai models: A comprehensive review. *Comput. Surveys* 57, 4 (2024), 1–35.
- [41] Pinlong Zhao, Weiyao Zhu, Pengfei Jiao, Di Gao, and Ou Wu. 2025. Data poisoning in deep learning: A survey. *arXiv preprint arXiv:2503.22759* (2025).

A Algorithmic Details of CACL

This appendix provides explicit pseudocode for the Confusability-Aware Contrastive Learning (CACL) framework. Algorithm 1 describes how pairwise class confusability weights are computed, and Algorithm 2 presents the full training loop integrating these weights into the contrastive regularization term.

B Additional Experimental Details

This section provides a detailed account of the experimental settings used in this paper to ensure full reproducibility. We detail the datasets, model architectures, and training hyperparameters.

B.1 Dataset Details

All datasets were used in their standard splits. For our generalizability study in Section 5.3.2, which involved CIFAR-100, we selected a representative subset of 10 classes to maintain visualization clarity while capturing diverse semantic relationships. The chosen source class was ‘beaver’, and the target classes included both semantically close classes (e.g., ‘otter’, ‘raccoon’) and dissimilar classes (e.g., ‘castle’, ‘rocket’).

Algorithm 1 Confusability Weights Precomputation

Input: Probe model f_{θ_0} , dataset $\mathcal{D}$, temperature $T > 0$, class set $\mathcal{Y} = \{1, \dots, C\}$

Output: Pairwise confusability weights $w(i, j)$ for all $i \neq j$

- 1: **Stage A: Centroids and Variances**
- 2: **for** $k \in \mathcal{Y}$ **do**
- 3: $\mathcal{D}_k \leftarrow \{(x, y) \in \mathcal{D} \mid y = k\}$
- 4: $\mathbf{Z}_k \leftarrow \{\mathbf{z} = f_{\theta_0}(x) \mid (x, k) \in \mathcal{D}_k\}$
- 5: $\mathbf{c}_k \leftarrow \frac{1}{|\mathbf{Z}_k|} \sum_{\mathbf{z} \in \mathbf{Z}_k} \mathbf{z}$ {class centroid}
- 6: $\text{var}_k \leftarrow \frac{1}{|\mathbf{Z}_k|} \sum_{\mathbf{z} \in \mathbf{Z}_k} \|\mathbf{z} - \mathbf{c}_k\|_2^2$
- 7: **end for**
- 8: **Stage B: Pairwise Confusability**
- 9: **for** $i \in \mathcal{Y}$ **do**
- 10: **for** $j \in \mathcal{Y}, j \neq i$ **do**
- 11: $C(i, j) \leftarrow \frac{\text{var}_i + \text{var}_j}{\|\mathbf{c}_i - \mathbf{c}_j\|_2^2 + \epsilon}$
- 12: **end for**
- 13: **end for**
- 14: **Stage C: Softmax Normalization**
- 15: **for** $i \in \mathcal{Y}$ **do**
- 16: $Z_i \leftarrow \sum_{j \neq i} \exp(C(i, j)/T)$
- 17: **for** $j \in \mathcal{Y}, j \neq i$ **do**
- 18: $w(i, j) \leftarrow \exp(C(i, j)/T)/Z_i$
- 19: **end for**
- 20: **end for**
- 21: **return** $w(\cdot, \cdot)$

B.2 Model Architectures

We used standard implementations for all network architectures.

- **ResNet-18:** We used the standard ResNet-18 architecture as implemented in PyTorch’s ‘torchvision.models’.
- **Vision Transformer (ViT):** We used the ViT-B/16 model, also from ‘torchvision.models’, with a patch size of 16x16 and an image size of 224x224 (CIFAR-10 images were upsampled accordingly).
- **Multi-Layer Perceptron (MLP):** Our MLP was designed for the 32x32 image size of CIFAR-10, flattened to a 3072-dimensional input vector. It consists of three hidden layers with 512, 256, and 128 neurons, respectively. Each hidden layer is followed by a ReLU activation function and Batch Normalization.

B.3 Hyperparameter Settings

All models were trained from scratch without the use of pre-trained weights. The key hyperparameters used for training on CIFAR-10 are summarized in Table 6. These settings were kept consistent across all methods unless otherwise specified by the original authors of the baseline methods. For our method, CACL, the balancing hyperparameter λ was set to 0.5, the contrastive temperature τ to 0.1, and the weighting temperature T to 1.0, as determined on a held-out validation set.

B.3.1 Implementation of Baselines. For all baseline methods (Sup-Con, TCL, SGN), we made our best effort to reproduce their results

Algorithm 2 Training with CACL

Input: Model f_{θ} , classifier g_{ϕ} , dataset $\mathcal{D}$, weights $w(i, j)$, batch size B , CACL weight $\lambda > 0$, temperature $\tau > 0$, momentum $m \in [0, 1)$, refresh period R

Output: Trained parameters θ, ϕ

- 1: Initialize class centroids $\{\mathbf{c}_k\}_{k=1}^C$
- 2: **for** epoch = 1, 2, ... **do**
- 3: **if** epoch mod $R = 0$ and dynamic weights enabled **then**
- 4: Recompute $w(i, j)$ via Alg. 1
- 5: **end if**
- 6: **for** minibatch $\mathcal{B} = \{(x_b, y_b)\}_{b=1}^B$ **do**
- 7: $\mathbf{z}_b \leftarrow f_{\theta}(x_b); \hat{\mathbf{p}}_b \leftarrow g_{\phi}(\mathbf{z}_b)$
- 8: $\mathcal{L}_{\text{CE}} \leftarrow \frac{1}{B} \sum_{b=1}^B \text{CE}(\hat{\mathbf{p}}_b, y_b)$
- 9: **CACL component:**
- 10: **for** $b = 1$ **to** B **do**
- 11: $i \leftarrow y_b; s(\mathbf{z}, \mathbf{c}) \leftarrow \exp(\cos(\mathbf{z}, \mathbf{c})/\tau)$
- 12: $N_b \leftarrow s(\mathbf{z}_b, \mathbf{c}_i)$
- 13: $D_b \leftarrow s(\mathbf{z}_b, \mathbf{c}_i) + \sum_{j \neq i} w(i, j) s(\mathbf{z}_b, \mathbf{c}_j)$
- 14: $\ell_b^{\text{CACL}} \leftarrow -\log(N_b/D_b)$
- 15: **end for**
- 16: $\mathcal{L}_{\text{CACL}} \leftarrow \frac{1}{B} \sum_{b=1}^B \ell_b^{\text{CACL}}$
- 17: $\mathcal{L} \leftarrow \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{CACL}}$
- 18: Update θ, ϕ by SGD/Adam on $\nabla_{\theta, \phi} \mathcal{L}$
- 19: **(Optional) EMA centroids update:**
- 20: **for** $k \in \mathcal{Y}$ **do**
- 21: $\tilde{\mathbf{c}}_k \leftarrow \text{mean}\{\mathbf{z}_b \mid y_b = k\}$ (if present)
- 22: **if** class k appears in batch **then**
- 23: $\mathbf{c}_k \leftarrow m \mathbf{c}_k + (1 - m) \tilde{\mathbf{c}}_k$
- 24: **end if**
- 25: **end for**
- 26: **end for**
- 27: **end for**
- 28: **return** θ, ϕ

faithfully. We either used the authors’ official open-source implementations or re-implemented them based on their papers within our unified experimental framework. Hyperparameters for these methods were set following the recommendations in their respective papers for CIFAR-10.

C Additional Experimental Results

In this section, we provide supplementary experimental results that further validate the findings presented in the main paper. We first demonstrate the generalizability of our method’s advantages across datasets with different characteristics (CIFAR-100 and Fashion-MNIST) in Section C.1. We then show that these advantages are robust to more severe attack scenarios with higher poison rates up to 40% in Section C.2.

C.1 Generalization to Other Datasets

To verify that the superior worst-case protection and robustness equity of CACL are not specific to CIFAR-10, we conducted identical experiments on the more complex, fine-grained CIFAR-100 dataset and the simpler, grayscale Fashion-MNIST dataset.

Table 4: Defense method performance on CIFAR-100 and Fashion-MNIST under a 10% poison rate.

		Targeted Attack (Worst-Case)		Random Attack		
Method	Clean Acc. (%)	Overall Acc. (%)	Worst-Class Acc. (%)	Overall Acc. (%)	Acc. Drop ↓	Acc. Std. Dev. ↓
CIFAR-100						
Standard CE	67.54	61.22	31.33	64.89	2.65	21.5
SupCon	68.11	64.56	42.18	66.01	2.10	16.3
TCL	67.80	64.31	41.50	66.15	1.65	15.8
SGN	68.25	64.63	42.35	66.28	1.97	15.5
CACL (Ours)	67.95	65.58	48.65	66.24	1.71	14.1
Fashion-MNIST						
Standard CE	92.15	88.30	82.50	90.55	1.60	4.5
SupCon	92.50	90.85	86.10	91.80	0.70	3.1
TCL	92.45	90.90	85.90	91.95	0.50	2.9
SGN	92.61	91.15	86.25	92.05	0.56	2.8
CACL (Ours)	92.55	91.55	87.80	91.98	0.57	2.4

Table 5: Defense method performance on CIFAR-10 under different poison rates.

		Targeted Attack (Worst-Case)		Random Attack		
Method	Clean Acc. (%)	Overall Acc. (%)	Worst-Class Acc. (%)	Overall Acc. (%)	Acc. Drop ↓	Acc. Std. Dev. ↓
<i>Poison Rate: 20%</i>						
Standard CE	79.53	69.80	55.20	71.30	8.23	14.5
SupCon	79.98	72.15	60.80	73.90	6.08	11.8
TCL	79.62	72.30	60.50	74.85	4.77	11.2
SGN	80.11	72.55	61.50	75.10	5.01	10.9
CACL (Ours)	79.95	74.90	68.10	74.95	5.00	9.5
<i>Poison Rate: 30%</i>						
Standard CE	79.53	65.10	44.30	67.50	12.03	19.8
SupCon	79.98	68.40	52.50	70.80	9.18	16.5
TCL	79.62	68.70	52.10	71.90	7.72	15.8
SGN	80.11	68.90	53.60	72.30	7.81	15.4
CACL (Ours)	79.95	72.10	60.25	72.15	7.80	13.1
<i>Poison Rate: 40%</i>						
Standard CE	79.53	60.30	35.50	63.80	15.73	24.2
SupCon	79.98	64.50	45.80	67.10	12.88	19.1
TCL	79.62	65.10	45.50	68.95	10.67	18.9
SGN	80.11	65.40	46.90	68.80	11.31	18.5
CACL (Ours)	79.95	69.20	55.10	68.65	11.30	15.9

The results, presented in Table 4, show that our core findings hold consistently. On the more challenging CIFAR-100 dataset, the performance of all methods degrades, but CACL’s unique strengths become even more apparent. While top methods like SGN and TCL are highly competitive in overall metrics, CACL provides a substantial boost in Worst-Class Accuracy (+6.3 points over the next best) and maintains the most equitable performance with the lowest Accuracy Std. Dev. Even on the simpler Fashion-MNIST dataset,

where absolute performance differences are smaller, the same trend holds. These results strongly suggest that the advantages of CACL stem from its fundamental, vulnerability-aware approach.

C.2 Performance under Different Poison Rates

We also investigate whether the observed advantages of CACL persist under more severe attack conditions. We repeated the CIFAR-10 experiments with higher poison rates of $p = 20\%$, 30% , and 40% .

Table 6: Key training hyperparameters for all models on CIFAR-10.

Hyperparameter	Value
Optimizer	SGD
Initial Learning Rate	0.1
Momentum	0.9
Weight Decay	5e-4
Learning Rate Schedule	Cosine Annealing
Batch Size	128
Training Epochs	100

The results are detailed in Table 5. As expected, the performance of all methods degrades as the attack intensity increases. Crucially, however, CACL’s superiority in protecting the most vulnerable class and maintaining robustness equity not only persists but becomes even more pronounced. Under the extreme 40% poison rate, the performance gap in Worst-Class Accuracy between CACL and the next best method widens to over 8 percentage points, showcasing its resilience when it matters most. This demonstrates that CACL is not only effective under moderate threat levels but also scales robustly to severe data poisoning scenarios, where its targeted approach provides a critical advantage.